**Biophysics 210: Biological Light Microscopy**
**Syllabus**

Discussion section meets Tuesdays from 1-2:30pm in GH N114
Labs meet Thursday or Friday from 1-4pm GH N114

**Week 8: Digital Images and Image Analysis**

**Goals:** Understand how information is encoded in digital images, what the bit depth of an image means, how images must be transformed for printing or display on a screen, and how RGB and monochrome images differ. Learn about basic image analysis operations. Use Fiji/ImageJ for executing basic image analysis operations.

**Discussion Section:** May $19^{th}$
**Labs:** May $21^{th}$ and May $22^{th}$

**Lectures** (watch before discussion section):
- Introduction to Digital Images
- Introduction to Bioimage Analysis

**Reading required for discussion section:**
- Waters, JC (2009) Accuracy and precision in quantitative fluorescence microscopy. J Cell Biol. 185:1135-1148
- Pawley, J (2000) The 39 steps: A cautionary tale of quantitative 3D fluorescence microscopy. BioTechniques 28(5):884-886
- Cromey, DW (2013) Digital Images Are Data: And Should Be Treated as Such. Methods Mol. Biol. 931:1-27

**Additional Reading** (optional):
*General*
- Dealing with Imaging Data
- Image Analysis (more in-depth video about using filters)
- Metadata in Bioimaging (Part of Bioimage Analysis Course on iBiology)
- Wait, EC et al (2020) Hypothesis-driven quantitative fluorescence microscopy- the importance of reverse-thinking in experimental design. J Cell Sci. 133:jcs250027
- Jost, APT and Water, JC (2019) Designing a rigorous microscopy experiment: Validating methods and avoiding bias. J Cell Biol 218:1452-1466
- Belthangady, C and Royer, LA (2019) Applications, promises, and pitfalls of deep learning for fluorescence image reconstruction. Nature Methods 16:1215-1225

*Open-Source Software Tutorials/Resources*

- The Scientific Community Image Forum (Image SC)
  - Forum for over 50 open access software platforms including ImageJ/Fiji, Cell Profiler, and napari. If you have an image analysis question this is the place to search for answers and post questions.
- Fiji
- Cell Profiler; also see Cell Profiler tutorials and Cell Profiler Blog
- napari; also see napari tutorials, napari-hub for sharing plugins, and GitHub repository
- Peter Bankhead: Introduction to Bioimage Analysis- Excellent primer for image analysis
- COBA: Center for Open Bioimage Analysis: YouTube channel with tutorials for ImageJ and Cell Profiler
- BioImaging North America: Training & Education Resources, Multiple great resources linked here.
- I2K meeting: From Images to Knowledge; links to all the talks and workshops from this year can be found on the website

*Papers on Useful Tools*


- Bray,MA. et al (2016) Cell painting, a high-content image-based assay for morphological profiling using multiplexed fluorescent dyes. Nature Protocols 11:1757-1774
- Ljosa V and Carpenter AE (2009) Introduction to the quantitative analysis of two-dimensional fluorescence microscopy images for cell-based screening. PLoS Comput Biol 5(12):e1000603
- Lucas, AM et al (2021) Open-source deep-learning software for bioimage segmentation. Mol Biol Cell 32:823-829
- Bolte, S and Cordelieres, FP (2010) A guided tour into subcellular colocalization analysis in light microscopy. J of Microscopy 224:213-232
- Sommer, C and Gerlich, DW (2013) Machine learning in cell biology- teaching computers to recognize phenotypes. J Cell Sci 126:5529-5539.
- Arganda-Carreras, I and Andrey, P (2017) Designing Image Analysis Pipelines in Light Microscopy: A Rational Approach. In: Markaki Y., Harz H. (eds) Light Microscopy. Methods in Molecular Biology, vol 1563. Humana Press, New York, NY

*Reporting, Reproducibility, and Ethics*

*Nature Methods Dec 2021 was an issue dedicated to imaging reproducibility discussing both how to keep your reporting accurate as well as tools that can be used! We are only highlighting a subset of references here.*

- Llopis, PM et al (2021) Best practices and tools for reporting reproducible fluorescence microscopy methods. Nat Methods 18:1463-1476
- Swedlow, JR et al (2021) A global view of standards for open image data formats and repositories. Nat Methods 18:1440-1446
- Boehm, U et al (2021) QUAREP-LiMi: a community endeavor to advance quality assessment and reproducibility in light microscopy. Nat Methods 18:1423-1426
- Moore, et al (2021) OME-NGFF: a next-generation file format for expanding bioimaging data-access strategies. Nat Methods 18:1496-1498

*Other papers on reproducibility and ethics*

- Lee, JY and Kitaoka M (2018) A beginner's guide to rigor and reproducibility in fluorescence imaging experiments. Mol Bio of the Cell https://doi.org/10.1091/mbc.E17-05-0276
- Aaron, J and Chew TL (2021) A guide to accurate reporting in digital image processing – can anyone reproduce your quantitative analysis? J Cell Sci 134:jcs254151
- Heddleston, JM et al (2021) A guide to accurate reporting in digital image acquisition – can anyone replicate your microscopy data? J Cell Sci 134:jcs254144
- Laine, RF et al (2021) Avoiding a replication crisis in deep-learning-based bioimage analysis. Nat Methods 18:1136-1144
- North, AJ (2006) Seeing is believing? A beginners' guide to practical pitfalls in image acquisition. J Cell Biol 172:9-18.
- Cromey, DW (2010) Avoiding Twisted Pixels: Ethical Guidelines for the Appropriate Use and Manipulation of Scientific Digital Images. Sci Eng Ethics 16:639-667
- Pearson, H (2007) The good, the bad and the ugly. Nature 447:138-140
- Pearson, H (2005) Image manipulation: CSI: cell biology. Nature 434:952-953

**Discussion Section Topic:** In the discussion section we will cover file formats, image analysis basics, common issues encountered, and how acquisition parameters impact your results.

**Lab:** In the lab we will work through a series of image analysis techniques and problems using the open-source Fiji distribution of ImageJ. Please install Fiji from http://fiji.sc/Fiji to your laptop before lab section and bring your laptop if you can. A link to the images for these exercises will be sent out with the lab section write up.